PUSHUP PRO

MVP Product Requirements Document - Version 3

Camera-based push-up counter with verified reps, AI goal coaching, and a Virtual Training Partner

Field	Decision
Product	PushUp Pro
MVP validation goal	Validate whether users trust and repeatedly use a camera-based push-up counter to track and total push-ups, while competing against a Virtual Training Partner.
Platform	Web app prototype built in Claude Code, Replit, Lovable, or similar tools.
Core interaction	User opens camera, app detects view angle, counts push-ups, saves verified reps, and updates personal progress.
Primary technical risk	Camera-based rep counting accuracy across front, side, and diagonal camera angles.
AI role	LLM-powered goal coach and Virtual Training Partner. Real-time counting is handled by the local vision pipeline, not the LLM.
MVP screens	Onboarding, Workout/Camera, Summary/Edit, Dashboard.

Product thesis

PushUp Pro should not start as a full fitness platform. The MVP exists to prove one hard thing: users trust the camera counter enough to use it repeatedly. AI, competition, and progress tracking are included only when they directly support that validation goal.

1. MVP Positioning and Product Boundary

1.1 MVP positioning statement

PushUp Pro MVP is a camera-based push-up counting web app that helps users track verified push-up reps, review progress, receive safe AI-generated goals, and compete against a Virtual Training Partner. The MVP does not attempt to be a complete bodyweight training platform. It is built to validate trust, repeat usage, and counting accuracy.

1.2 Target users for MVP

- Beginners trying to build consistency at home.
- Students and young adults who want a simple, motivating home workout tool.
- Athletic users who want verified rep totals and a lightweight competitive loop.

1.3 Non-goals for MVP

- No full bodyweight trainer experience in MVP.
- No form correction beyond basic setup and rep-state guidance.
- No real friend network. Competition is against mocked bot competitors and one Virtual Training Partner.
- No native app, wearable integrations, subscriptions, real payment system, or real push notifications.
- No video recording, image uploads, or sharing camera content.

2. Minimal MVP Scope

MVP item	Included?	Decision
Camera push-up counting	Yes	Core feature. Must support front, side, and diagonal views using local pose detection and orientation-specific rules.

Manual rep entry	Yes	Fallback only. Manual reps count toward personal progress but are labelled unverified.
AI goal coach	Yes	Real LLM integration using structured workout data, safety constraints, and user inputs.
Virtual Training Partner	Yes	AI-created competitor that uses simulated progress and supportive comparison.
Bot challenges	Yes	Mocked competitors only. Challenge target is achievable and personalized, not fixed at 500.
Smart reminders	Limited	In-app reminder cards and LLM-generated reminder copy only. No push notifications in MVP.
Badges/streaks	Very limited	Only dashboard indicators needed to support repeat usage. No full badge system or badge screen in MVP.
Push-up variations	Limited	Explicitly labelled by support level: camera-supported, manual-only, educational-only, future support.
Ads/monetization	Experiment only	No disruptive ads during active sets. Use post-set sponsor cards only if needed for test.

3. Required MVP Screens

Screen	Purpose	Must include
1. Onboarding	Collect enough data for safe goals and camera setup.	Name, fitness level, current max reps, weekly availability, soreness/fatigue check, manual goal optional, consent to camera processing and AI coach.
2. Workout/Camera	Run camera setup, classify camera angle, count reps, display verified count.	Camera preview, camera-on indicator, orientation label, readiness status, rep counter, set timer, end set, fallback prompts.
3. Summary/Edit	Confirm workout result and separate verified from unverified data.	Detected reps, editable final reps, verified/unverified label, camera confidence, notes, pain/fatigue check, save button.
4. Dashboard	Show progress and motivate next action.	Today reps, weekly reps, verified reps, unverified reps, next AI goal, Virtual Training Partner status, in-app reminder card.

4. Technical Architecture for Camera Counting

4.1 Architecture decision

Key decision

The LLM must not perform real-time frame-by-frame rep counting. Real-time counting is handled by the local computer vision pipeline because it is faster, cheaper, more private, and more reliable. The LLM receives structured pose summaries, workout history, safety inputs, and optional setup snapshots only when the user consents.

4.2 System components

Component	Responsibility	Implementation guidance

Browser camera layer	Requests camera permission, shows live preview, confirms camera-on state.	Use getUserMedia. Never access camera in background. Stop tracks when workout ends.
Local pose detection layer	Detects body landmarks and confidence values from camera frames.	Use MediaPipe Pose, TensorFlow.js pose detection, or equivalent. Process in browser when possible.
View-angle classifier	Classifies camera angle as front, side, diagonal, or unsupported.	Uses landmark visibility and shoulder/hip geometry. Must output angle and confidence.
Rep-state machine	Counts reps using orientation-specific motion profiles.	States: SETUP_READY, TOP, DESCENDING, BOTTOM, ASCENDING, REP_COMPLETE, LOST_TRACKING.
Quality gate	Decides whether current tracking is reliable enough to count verified reps.	Requires minimum landmark confidence, stable body framing, and consistent state transitions.
LLM coach layer	Generates safe goals, summaries, reminders, and Virtual Training Partner messages.	Consumes structured data only. No raw video storage. Optional single-frame setup analysis may be enabled with explicit consent.
Analytics/calculation layer	Centralizes totals, streaks, verified counts, challenge status, and goal inputs.	All dashboard numbers must come from this layer, not separate page-level calculations.

4.3 Camera angle support

The MVP must support three camera views: front view, side view, and diagonal view. The app must first classify the view, then apply the correct rep-counting profile. If confidence is too low, the app must guide the user to reposition instead of guessing.

Camera view	How the app identifies it	Rep-counting signal	Common failure risk
Front view	Both shoulders and both hips visible with roughly symmetrical left/right landmark positions.	Vertical body movement, shoulder/hip height change, elbow bend pattern, top-to-bottom-to-top cycle.	Depth is harder to judge; app should be more forgiving but require clear up/down movement.
Side view	One side of body dominates; shoulder, hip, elbow, knee/ankle visible along a horizontal body line.	Elbow angle, shoulder-to-hip alignment, body descent and return, clearer depth estimate.	If body is partly off-frame, elbow/hip confidence may drop.
Diagonal view	Both sides partially visible, but one side has stronger landmark confidence.	Hybrid of side and front profile. Use dominant-side elbow angle plus vertical movement.	More occlusion and inconsistent depth; requires higher confidence threshold than side view.
Unsupported	Multiple people, too dark, body not visible, only face visible, or landmarks below threshold.	No verified counting.	Must show reposition instructions and allow manual unverified entry.

4.4 Rep-state machine

State	Definition	Transition condition
SETUP_READY	User is visible, view angle is classified, confidence threshold is met.	User holds top position for 1 second or taps Start Set.
TOP	Arms extended enough for starting position based on selected view.	Elbow/body movement indicates controlled descent.
DESCENDING	User is lowering from top position.	Body reaches bottom threshold or descent stalls.
BOTTOM	User reaches enough depth for a forgiving valid rep.	User begins upward movement.
ASCENDING	User returns toward top position.	User reaches top threshold again.
REP_COMPLETE	One full cycle has been completed.	Increment verified rep count, return to TOP.
LOST_TRACKING	Landmarks confidence or body framing drops below threshold.	Pause verified counting and show recovery prompt.

4.5 Accuracy acceptance criteria

Test condition	Acceptance target
Controlled lighting, supported camera view, standard push-ups	Average error within +/-2 reps over a 20-rep set.
Front view, supported setup	At least 80% of test sessions save without manual correction.
Side view, supported setup	At least 85% of test sessions save without manual correction.
Diagonal view, supported setup	At least 80% of test sessions save without manual correction.
Unsupported camera setup	App must not count verified reps. It must enter LOST_TRACKING or SETUP_NOT_READY and guide the user.
False positive prevention	App must not count more than 1 false rep during a 30-second non-push-up movement test.
Manual correction rate	80% of camera sessions should be saved without manual correction in MVP test group.

5. LLM Requirements and Prompting

5.1 What the LLM can and cannot do

Area	Allowed in MVP	Not allowed in MVP
Goal coaching	Generate daily and weekly push-up goals from workout history, fitness level, fatigue, soreness, and manual goals.	Do not recommend unsafe volume jumps or medical advice.
Camera understanding	Read structured pose summaries and optional setup snapshot descriptions after user consent.	Do not directly count every camera frame or store raw video.
Virtual Training Partner	Generate a supportive AI competitor with simulated progress and motivational messages.	Do not shame, insult, diagnose, or pressure a user who reports pain.
Reminders	Generate in-app reminder card copy based on progress and safety state.	No actual push notifications in MVP. No pain-ignoring streak pressure.

5.2 LLM inputs

- User profile: fitness level, current max reps, goal preference, weekly availability, training age if provided.
- Workout history: verified reps, unverified reps, daily totals, weekly totals, best set, average reps per set, missed days.
- Current session: detected camera angle, camera confidence score, verified reps, manual edits, pain/fatigue report.
- Safety state: beginner cap, soreness status, pain flag, heavy-volume flag, recent volume increase percentage.
- Manual goals: user-specified daily or weekly target if provided.
- Competition status: Virtual Training Partner target, bot challenge progress, user gap to target.

5.3 Overtraining prevention rules

Rule	MVP requirement
Beginner weekly cap	For Beginner level, AI may not increase weekly target by more than 10% from the previous completed week unless user manually overrides.
Intermediate weekly cap	For Intermediate level, default increase limit is 15%.
Advanced/Athlete cap	Default increase limit is 20%, unless pain/fatigue is reported.
Pain flag	If user reports pain, AI must recommend stopping, resting, or lowering volume. It must not issue competitive pressure.
Heavy volume flag	If weekly volume increases by more than cap, dashboard must show a recovery warning and suggest a lighter day.
Missed workout logic	AI must not force users to make up all missed reps in one day. It should offer a safe adjusted goal.

5.4 Exact LLM prompt templates

Prompt type	System/developer instruction	Example user data	Expected response
Daily goal	You are PushUp Pro, a supportive fitness coach. Use only the provided data. Create a safe push-up goal. Never provide medical advice. Respect the safety caps.	Fitness: Beginner. Max set: 12. Last week verified reps: 70. Missed workouts: 2. Soreness: mild. Manual goal: none.	Today, aim for 10-15 quality push-ups across 2-3 small sets. Keep it controlled and stop if soreness turns into pain.
Workout summary	Summarize the workout in 2 sentences. Separate verified and unverified reps. Encourage the user without exaggeration.	Verified reps: 22. Manual unverified reps: 5. Camera angle: side. Best set: 12. Prior best: 10.	Strong session. You completed 22 verified reps and added 5 unverified reps manually. You also beat your best set with 12 reps.
Virtual Training Partner	Create a supportive competitor message. Motivate without shaming. If pain is reported, prioritize rest and safety.	User weekly verified reps: 64. Partner weekly reps: 72. Pain flag: false. Challenge target: 100.	Your Virtual Training Partner is only 8 reps ahead. One controlled set today puts you right back in the race.
Pain state reminder	Safety overrides competition. Do not mention streak pressure or leaderboard pressure when pain is reported.	Pain flag: true. Weekly target: 100. Current reps: 80.	You reported pain, so today is not about chasing reps. Rest, recover, and only return when movement feels comfortable.
Camera setup help	Explain the camera issue in plain language and give one action to fix it.	Detected view: unsupported. Reason: upper body only, hips not visible. Lighting: good.	I can see your upper body, but I need your hips in frame to count reps. Move the camera back slightly before starting.

6. Competition, Verified Reps, and Push-Up Variations

6.1 Virtual Training Partner naming decision

Use the name Virtual Training Partner instead of AI Clone or AI Rival. It is less creepy, more supportive, and better aligned with the beginner-friendly MVP. The partner can still behave competitively, but the default framing is accountability, not surveillance or intimidation.

6.2 Bot challenge design

Challenge change

Replace the fixed First to 500 challenge with a personalized Weekly Target Race. The user competes against a Virtual Training Partner and mocked bot competitors toward a target generated from their fitness level and history.

Fitness level	Default week 1 challenge target	Notes
Beginner	40-100 verified reps/week	Target depends on current max reps and soreness.
Intermediate	100-200 verified reps/week	Can include manual unverified reps for personal progress but not challenge ranking.
Advanced	200-350 verified reps/week	Progress target may increase if accuracy and retention are strong.
Athlete	350-500 verified reps/week	500 is reserved for Athlete-level users only, not default beginners.

6.3 Verified vs. unverified reps

Rep type	Counts toward personal progress?	Counts toward challenges?	Displayed how?
Verified camera-counted reps	Yes	Yes	Shown as Verified Reps.
Manual reps added after camera session	Yes	No	Shown as Unverified Reps.
Manual-only workout reps	Yes	No	Shown as Personal Progress only.
Rejected/uncertain camera reps	No by default; user may add manually as unverified.	No	Shown in summary as Needs Review.

6.4 Variation support labels

Variation	MVP support label	Reason
Standard push-up	Camera-supported in MVP	Primary supported movement.
Knee push-up	Camera-supported in MVP if side/diagonal view confidence is high	Useful for beginners, but depth rules differ.
Incline push-up	Manual-only in MVP	Surface height changes angle and depth logic.
Wide-grip push-up	Educational-only in MVP	Can distort elbow angle assumptions.
Diamond push-up	Educational-only in MVP	Hand position changes elbow/shoulder tracking.
Decline push-up	Future camera support	Requires separate body-angle logic.
Archer/explosive/clap/one-arm progressions	Future camera support	Too complex for MVP counting accuracy.

7. Monetization Strategy

7.1 Recommended strategy

Monetization decision

Do not place ads during active sets. The best MVP monetization direction is a free athlete training tool supported by non-intrusive sponsor placements and future team/athlete premium features. Active workout screens must stay distraction-free.

Monetization option	MVP decision	Reason
Ads during active set	Not allowed	Distracts users, harms safety, and weakens premium athletic feel.
Bottom/banner ads on camera screen	Not recommended	Could clutter the most important screen and reduce trust.
Post-set sponsor card	Allowed as experiment only	Appears after the set, not during movement. Must be skippable and fitness-relevant.
Dashboard sponsor tile	Allowed as experiment only	Can support free usage without interrupting training.
Premium individual plan	Future	Advanced form coaching, deeper analytics, and custom programs.
Team/athlete plan	Best long-term fit	Position PushUp Pro as an athlete training and accountability tool for teams, coaches, schools, and camps.

7.2 Long-term athlete training monetization

- Free user plan: camera counting, verified reps, basic goals, dashboard, and Virtual Training Partner.
- Team plan: coach dashboard, athlete leaderboards, assigned challenges, verified rep reports, team accountability groups.
- Sponsored challenge model: brands sponsor optional challenges, but the workout itself remains distraction-free.
- Premium individual plan: advanced form feedback, custom programs, voice coaching, and exportable progress reports after retention is proven.

8. Fallback States, Safety, and Privacy

8.1 Camera fallback states

State	Trigger	User-facing message	System action
CAMERA_PERMISSION_DENIED	User blocks camera access.	Camera access is needed for	Show permission instructions and

		verified reps. You can still enter reps manually as unverified.	manual entry option.
NO_PERSON_DETECTED	No pose landmarks above threshold.	I cannot see you yet. Step into frame and make sure your full body is visible.	Do not count. Keep camera preview active.
PARTIAL_BODY_VISIBLE	Key landmarks missing, often hips/shoulders.	I need your shoulders and hips in frame to count accurately. Move the camera back slightly.	Do not count verified reps.
LOW_LIGHT	Pose confidence unstable due to lighting.	Lighting is too low for verified counting. Move to brighter light or use manual entry.	Pause verified counting.
UNSUPPORTED_ANGLE	Angle classifier confidence below threshold.	I cannot tell your camera angle. Try side view for the most reliable counting.	Guide setup; no verified reps.
LOST_TRACKING_MID_SET	Landmark confidence drops during set.	Tracking was lost. Hold still at the top position to resume.	Pause counting; preserve current verified count.
MULTIPLE_PEOPLE_DETECTED	More than one body detected.	I see more than one person. Please train alone in frame for verified counting.	Pause counting.
FAST_OR_PARTIAL_REPS	Movement is too fast or state transition incomplete.	Some reps may not be counted. Slow down and reach the full top position.	Count only valid cycles; allow manual unverified correction after set.

8.2 Safety UX requirements

- Warm-up prompt must appear before workout: "Warm up your shoulders, wrists, chest, and core before starting."
- Rest-day recommendation must appear after heavy volume or soreness input.
- Beginner weekly goal caps must be enforced by AI goal logic.
- Volume warning must appear if weekly target increases faster than the allowed cap.
- User can pause streak without shame. Use copy like: "Recovery day logged. Your progress still counts."
- Reminder copy must be safety-first. Never pressure users who report pain.
- Do not show "do not break the streak" messages if pain flag is true.

8.3 Privacy requirements

- No video recording in MVP.
- Camera stream processed locally when possible.
- Clear camera-on indicator must be visible whenever camera is active.
- Permission request must explain why camera access is needed.
- No background camera access.
- No image uploads in MVP.
- No sharing camera content with friends, bots, or Virtual Training Partner.
- Data deletion option must exist in settings or dashboard.
- Stored workout data: profile inputs, verified rep totals, unverified rep totals, session timestamps, camera angle label, confidence score, goals, and user safety inputs. Raw video is not stored.

9. MVP Data Model and Central Analytics Layer

9.1 Data model

Object	Fields
UserProfile	userId, name, fitnessLevel, currentMaxReps, weeklyAvailabilityDays, manualGoal, consentAccepted, createdAt, updatedAt
SafetyCheck	checkId, userId, date, sorenessLevel, painFlag, fatigueLevel, recoveryDayLogged
WorkoutSession	sessionId, userId, startedAt, endedAt, selectedVariation, cameraAngle, angleConfidence, trackingQuality, verifiedReps, unverifiedReps, originalDetectedReps, finalSavedReps, manuallyEdited, notes
RepEvent	repId, sessionId, repNumber, timestamp, startState, bottomState, endState, confidence, countedAsVerified
Goal	goalId, userId, type, targetVerifiedReps, startDate, endDate,

	generatedBy, safetyCapApplied, status
VirtualTrainingPartner	partnerId, userId, name, tone, weeklyTarget, currentSimulatedReps, messageHistory, lastUpdated
Challenge	challengeId, userId, name, targetVerifiedReps, startDate, endDate, botParticipants, userVerifiedReps, status
ReminderCard	reminderId, userId, createdAt, message, triggerReason, safetyState, dismissed

9.2 Central analytics functions

Function	Input	Output / rule
calculateDailyStats()	WorkoutSession[]	todayVerifiedReps, todayUnverifiedReps, totalSessions
calculateWeeklyStats()	WorkoutSession[]	weeklyVerifiedReps, weeklyUnverifiedReps, trainingDays
calculateBestSet()	WorkoutSession[] / RepEvent[]	best verified set only; unverified reps excluded.
calculateTrackingQuality()	angleConfidence, landmarkConfidence, lostTrackingCount	HIGH, MEDIUM, LOW, or FAILED.
evaluateGoalSafety()	profile, history, safetyCheck, proposedGoal	approvedGoal, capApplied, safetyWarning.
calculateChallengeProgress()	verified reps only	percentComplete and rank vs bots.
generateCoachContext()	profile, stats, safety, challenge	Structured JSON sent to LLM.

10. QA Requirements and MVP Experiment Plan

10.1 QA test matrix

Test area	Required cases	Pass condition
Camera permission	Allow, deny, revoke permission.	App handles all states without crashing.
Camera angle classification	Front, side, diagonal, unsupported.	Correct label in at least 85% of controlled setup tests.
Lighting	Bright room, dim room, backlit room.	Counts only when confidence threshold is met; otherwise shows fallback.
Body framing	Full body, upper-body only, partial hips, user leaves frame.	Verified counting pauses when required landmarks are missing.
Rep speed	Slow reps, normal reps, fast reps.	Normal reps meet accuracy target; fast uncertain reps prompt slowdown.
Partial reps	Half reps, shallow reps, top-only movement.	No more than 1 false rep in 30-second partial movement test.
Manual correction	Edit count up/down after set.	Manual changes are saved as unverified and excluded from challenges.
Safety checks	Pain true, soreness high, heavy volume.	AI and reminders switch to safety-first copy.
Data deletion	Delete progress data.	Stored profile/workout data is removed from local storage or configured database.

10.2 Real-user MVP experiment plan

Phase	Participants	What to test	Decision criteria
Internal alpha	3-5 builders/testers	Camera access, orientation detection, rep count flow, basic dashboard.	No critical crashes; camera flow works on target browsers.
Controlled accuracy test	8-10 users, mixed fitness levels	Each user performs 3 sets of 20 reps from front, side, and diagonal views.	Average error within +/-2 reps over 20-rep sets in controlled conditions.
7-day retention test	15-25 students/young adults	Use app at home for one week with Virtual Training Partner and in-app reminders.	40% return within 7 days; 70% complete first workout.
Challenge test	Same group	Prompt users to join personalized Weekly Target Race	30% join challenge when prompted.

		against bots.	
Trust survey	All beta users	Ask whether users trusted count, edited reps, and would use again.	70% say they would use again; manual correction reason is logged.

10.3 MVP success criteria

- 70% of onboarded users complete one workout.
- 40% of onboarded users return within 7 days.
- 80% of camera sessions save without manual correction.
- Rep counter averages within +/-2 reps over a 20-rep set in controlled tests.
- 30% of users join a challenge when prompted.
- Zero raw video files are stored during MVP testing.

11. Responsibility Matrix

Domain	Owner	Responsibilities
Product	Product Lead	MVP scope, acceptance criteria, roadmap gates, user experiment decisions.
Computer vision	CV/Frontend Engineer	Camera access, pose detection, view classification, rep-state machine, tracking quality.
Frontend/UI	Frontend Engineer + Designer	Four MVP screens, camera UX, fallback states, dashboard clarity, accessibility.
AI/LLM	AI Engineer / Prompt Owner	LLM prompts, context schema, safety rules, goal logic, Virtual Training Partner messages.
Data/analytics	Data Owner	Central analytics functions, event schema, success metrics, experiment reporting.
Safety/privacy	Product + Legal/Compliance Reviewer	Safety copy, pain-state rules, consent language, no-video-storage policy.
QA	QA Owner	Test matrix execution, browser/device coverage, accuracy validation, bug triage.

12. Claude Code / Replit / Lovable Build Sequence

1. Validate browser camera access using getUserMedia and show a clear camera-on indicator.
2. Validate basic pose/rep-counting feasibility using a local pose detection library or mockable pose service.
3. Build the view-angle classifier for front, side, diagonal, and unsupported setups.
4. Define and implement camera setup constraints and fallback states before building polish.
5. Implement the rep-state machine and verified rep counter.
6. Build the four MVP screens: Onboarding, Workout/Camera, Summary/Edit, Dashboard.
7. Implement verified vs. unverified rep handling and save workout sessions.
8. Create the central analytics/calculation layer and connect dashboard values to it.
9. Integrate the LLM for safe goals, summaries, reminder cards, and Virtual Training Partner messages.
10. Add the personalized Weekly Target Race against mocked bot competitors.
11. Run QA test matrix and controlled user experiment before adding any new feature surfaces.

13. Roadmap with Decision Gates

Future feature	Only build if...	Reason
Form feedback	Rep-count accuracy reaches acceptance target across supported views.	Form feedback is not credible if counting is not trusted.
Real social/friends	7-day retention and challenge join rate	Social features add complexity and

	show clear engagement.	privacy risk.
New exercises	Data model supports exercise-specific rules and push-up MVP is stable.	Each exercise has unique counting logic.
Premium plan	Users demonstrate repeated engagement and willingness to pay for advanced coaching.	Do not monetize before usage value is proven.
Team/athlete product	Coaches or teams show interest in verified rep reports and group challenges.	This is the strongest long-term business direction.

14. Final MVP Summary

PushUp Pro MVP is a focused web app prototype that validates camera-based push-up counting, verified progress tracking, and AI-supported accountability. The product supports front, side, and diagonal camera views through a local computer vision pipeline, then uses an LLM to generate safe goals, summaries, reminder cards, and Virtual Training Partner messages from structured workout data. The MVP is intentionally limited to four screens and excludes full form coaching, real social networking, native mobile development, video storage, and disruptive ads.

Build-ready one-line instruction

Build a four-screen web app where users onboard, complete a camera-counted push-up session, review verified/unverified reps, and return to a dashboard that shows progress, a safe AI goal, and a Virtual Training Partner challenge.